Orders

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> There are three primary concerns in life: what to eat for breakfast, lunch, and dinner.
>
> MahjongSoul story

My attempt at presenting orders. You should already know how to do basic computations modulo n.

1 Background

I think it's best if you know what a group is.

Definition 1.1. A *group* is an ordered pair $(G, \star)$ of a set G and a binary operation $\star$ such that:

- The operation $\star$ is associative: for all $a, b, c \in G$, we have $(a \star b) \star c = a \star (b \star c)$.
- There exists an identity. That is, there is an element $e \in G$ such that for all $a \in G$, $a \star e = e \star a = a$.
- Every element in G has an inverse. That is, for all $a \in G$, there exists an inverse a^{-1} such that $a \star a^{-1} = a^{-1} \star a = e$.

Exercise 1.2. Show that the identity and inverses are unique.

Definition 1.3. We define $\mathbb{Z}/n\mathbb{Z}$ to be the residue classes $\pmod n$, and $(\mathbb{Z}/n\mathbb{Z})^\times$ to be the set of residue classes $\pmod n$ relatively prime to n. For example,

$$\mathbb{Z}/20\mathbb{Z} = \{0, 1, \dots, 19\} \text{ and } (\mathbb{Z}/20\mathbb{Z})^\times = \{1, 3, 7, 9, 11, 13, 17, 19\}.$$

Here are some examples of groups.

- $(\mathbb{Z}, +)$: it has identity 0 and the inverse of n is $-n$.
- $(\mathbb{Z}/n\mathbb{Z}, +)$: it has identity 0 and the inverse of x is $-x \pmod n$.
- $(\mathbb{Q} \setminus \{0\}, \cdot)$: it has identity 1 and the inverse of x is $\frac{1}{x}$.
- $((\mathbb{Z}/20\mathbb{Z})^\times, \cdot)$: it has identity 1, and the inverses of $1, 3, 7, 9, 11, 13, 17, 19$ are $1, 7, 3, 9, 11, 17, 13, 19$, respectively. In fact, $((\mathbb{Z}/n\mathbb{Z})^\times, \cdot)$ is always a group.

Here are some non-examples of groups.

- $(\mathbb{R}, \cdot)$, since 0 has no inverse.
- $(\mathbb{Z}/20\mathbb{Z}, \cdot)$, since $0, 2, 4, 5, 6, 8, 10, 12, 14, 15, 16, 18$ all have no inverse. (Can you see why?)

Remark. If $\star$ is clear contextually, we sometimes denote the group by just the set G.

Now, how can we tell if two groups are the same?

Definition 1.4. A *isomorphism* between two groups $(G, \star)$ and $(H, \circ)$ is essentially a structure-preserving bijection. That is, a function $f : G \to H$ is an isomorphism if:

- f is bijective.
- For all $g_1, g_2 \in G_1,\ f(g_1 \star g_2) = f(g_1) \circ f(g_2)$.

If there exists an isomorphism between $(G, \star)$ and $(H, \circ)$, we say $(G, \star)$ and $(H, \circ)$ are *isomorphic*, denoted $(G, \star) \cong (H, \circ)$.

Remark. Isomorphism is an equivalence relation, that is, if G, H, I are groups, then:

- $G \cong G$,
- if $G \cong H$, then $H \cong G$,
- and if $G \cong H$ and $H \cong I$, then $G \cong I$.

If two groups are isomorphic, they are basically "the same thing with different names". For example, $(\mathbb{Z}, +) \cong (10\mathbb{Z}, +)$ where $10\mathbb{Z}$ is the set of multiples of 10 with the isomorphism $f(x) = 10x$. Also, $(\mathbb{R}^+, \cdot) \cong (\mathbb{R}, +)$ with the isomorphism $f(x) = \log x$ (any base), since

$$\log(a \cdot b) = \log(a) + \log(b).$$

Group isomorphisms are important since if we understand a group, we automatically understand all isomorphic groups.

Definition 1.5. The *direct product* of two groups $(G, \star)$ and $(H, \circ)$, denoted $(G, \star) \times (H, \circ)$, is the group defined on the set $G \times H$ and the operation $*$ defined by

$$(g_1, h_1) * (g_2, h_2) = (g_1 \star g_2, h_1 \circ h_2).$$

We use exponents to denote a group's direct product with itself.

For example, $((\mathbb{Z}/8\mathbb{Z})^\times, \cdot) \cong (\mathbb{Z}/2\mathbb{Z}, +)^2$ under the isomorphism f, where

$$f(1) = (0, 0),\ f(3) = (1, 0),\ f(5) = (0, 1), \text{ and } f(7) = (1, 1).$$

Exercise 1.6. Verify that f above is indeed an isomorphism.

Exercise 1.7. Verify the direct product of two groups is indeed a group.

Now we have all the tools we need to talk about number theory.

2 First steps

Our goal here is to understand the structure of the group $((\mathbb{Z}/p\mathbb{Z})^\times, \cdot)$, which happens to be surprisingly simple. First, we must actually verify that this is a group. Since the identity is 1 and multiplication is associative, we just need to show inverses exist:

Theorem 2.1 (Inverses exist)**.** For all $a \in (\mathbb{Z}/p\mathbb{Z})^\times$, there exists a unique element $b \in (\mathbb{Z}/p\mathbb{Z})^\times$ (commonly denoted a^{-1}) such that $ab \equiv 1 \pmod p$.

Proof. We claim the set equivalence

$$\{a, 2a, \dots, (p-1)a\} = \{1, 2, \dots, p-1\},$$

which implies the conclusion. Clearly $a, 2a, \dots, (p-1)a$ are all nonzero $\pmod p$. We claim they are also distinct, which finishes the proof by pigeonhole. Suppose $ma \equiv na \pmod p$ with $m < n$. But this implies $(n-m)a \equiv 0 \pmod p$, or $p \mid a$ or $p \mid n - m$, impossible since $0 < a, n - m < p$. □

We can now prove the famous Fermat's little theorem:

Theorem 2.2 (Fermat's little theorem)**.** For any $a \in (\mathbb{Z}/p\mathbb{Z})^\times$, we have

$$a^{p-1} \equiv 1 \pmod p.$$

Proof. Since

$$\{a, 2a, \dots, (p-1)a\} = \{1, 2, \dots, p-1\},$$

$(a)(2a)\dots((p-1)a) \equiv (1)(2)\dots(p-1) \pmod p$. Therefore

$$a^{p-1}(p-1)! \equiv (p-1)! \pmod p,$$

which implies the conclusion. □

Example 2.3. Compute $3^{1000} \pmod{23}$.

From Fermat's little theorem, we have

$$3^{1000} \equiv 3^{22\cdot 45+10} \equiv (3^{22})^{45} \cdot 3^{10} \equiv 3^{10} \equiv 2 \pmod{23}.$$

Now it is time to introduce the notion of orders.

Definition 2.4. The *order* of an element $a \in (\mathbb{Z}/p\mathbb{Z})^*$, denoted $\operatorname{ord}_p(a)$, is the smallest m such that $a^m \equiv 1 \pmod p$.

Theorem 2.5 (Fundamental theorem of orders)**.** If $a^n \equiv 1 \pmod p$, then $\operatorname{ord}_p(a) \mid n$.

Proof. Suppose otherwise, then write $n = q \operatorname{ord}_p(a) + r$ where $0 < r < \operatorname{ord}_p(a)$. Now

$$a^n \equiv \left(a^{\operatorname{ord}_p(a)}\right)^q \cdot a^r \equiv a^r \equiv 1 \pmod p,$$

but this is impossible since $r < \operatorname{ord}_p(a)$. □

Example 2.6 (AIME I 2019/14)**.** Find the least odd prime factor of $2019^8 + 1$.

Suppose $p \mid 2019^8 + 1$, now we see $2019^8 \equiv -1 \pmod p$, therefore $2019^{16} \equiv 1 \pmod p$. This implies $\operatorname{ord}_p(2019) = 16$ since the order divides 16 but not 8. But by Fermat's little theorem we have $2019^{p-1} \equiv 1 \pmod p$, so $16 \mid p - 1$ or $p \equiv 1 \pmod{16}$. Testing the second $1 \pmod{16}$ prime 97 works.

3 Primitive roots

Having introduced orders, we are now ready for primitive roots. The notion of a primitive root allows us to turn exponentiation and multiplication, which are generally hard to understand, into multiplication and addition, which are comparatively easier to understand. Therefore, the notion of a primitive root is critical to our understanding of the integers modulo p.

Definition 3.1. A *primitive root* g modulo p is an element in $(\mathbb{Z}/p\mathbb{Z})^\times$ with order $p-1$. In other words,

$$g^1, g^2, \ldots, g^{p-2}$$

are all not congruent to 1 modulo p.

This definition might seem a bit arbitrary. However, primitive roots have the following important property:

Theorem 3.2 (Primitive roots generate $(\mathbb{Z}/p\mathbb{Z})^\times$)**.** For all primitive roots g modulo p, we have the set equivalence

$$(\mathbb{Z}/p\mathbb{Z})^\times = \{1, 2, \ldots, p-1\} = \{g^0, g^1, \ldots, g^{p-2}\}.$$

Proof. The only values g^m can ever take are $1, 2, \ldots, p-1$. Therefore, we are done if we prove $g^0, g^1, \ldots, g^{p-2}$ are pairwise distinct, since this implies they take the values $1, 2, \ldots, p-1$ exactly once by pigeonhole. Now suppose by way of contradiction $g^a \equiv g^b \pmod{p}$ with $a < b$. Now $g^{b-a} \equiv 1 \pmod{p}$, with $1 \le b-a \le p-2$, contradiction since g is a primitive root. □

In general, it's hard to *find* primitive roots, but we have the following (very useful) existence theorem, whose proof we omit.

Theorem 3.3. There exist primitive roots modulo *every prime*.

This is incredibly useful. In fact, this is enough to imply the following:

Theorem 3.4. We have

$$((\mathbb{Z}/p\mathbb{Z})^\times, \cdot) \cong (\mathbb{Z}/(p-1)\mathbb{Z}, +).$$

Proof. Pick any primitive root g. We claim $f(x) \equiv g^x \pmod{p}$ is an isomorphism. We see f is bijective since $\{g^0, g^1, \ldots, g^{p-2}\} = \{1, \ldots, p-1\}$ by Theorem 3.2, and for any $a, b \in \mathbb{Z}/(p-1)\mathbb{Z}$, $g^{a+b} = g^a \cdot g^b$, so this is indeed an isomorphism. □

Invoking Theorem 3.4 to turn multiplication into addition is very powerful. This theorem basically says that, instead of dealing with multiplication modulo p, we can instead deal with addition modulo $p-1$.

Let's see a few examples. I strongly encourage you to try the examples yourself before reading the solutions.

Example 3.5 (Weakest HMMT Guts 2019/28)**.** For how many integers a with $1 \le a \le 41$ is the sequence $a^1, a^2, a^4, a^8, \ldots$ eventually constant $\pmod{41}$?

We see 41 works, now we count the $a \in (\mathbb{Z}/41\mathbb{Z})^\times$. Let $a = g^k$. We want the sequence $g^k, g^{2k}, g^{4k}, \ldots$ to eventually converge, which is equivalent to the exponents $k, 2k, 4k, \ldots$ converging $\pmod{40}$. This occurs iff k is a multiple of 5, of which there are 8. Therefore, the asnwer is 9.

Example 3.6 ($n^2 + 1$ lemma)**.** Let p be a prime. Then there exists an n such that $p \mid n^2 + 1$ iff $p \equiv 1 \pmod 4$.

Note the condition $p \mid n^2+1$ rearranges to $n^2 \equiv -1 \pmod p$, which becomes $\operatorname{ord}_p(n) = 4$. If $p \equiv 1 \pmod 4$, then taking $n = g^{\frac{p-1}{4}}$ where g is a primitive root works. If $p \mid n^2+1$, $\operatorname{ord}_p(n) = 4 \mid p-1$ so $p \equiv 1 \pmod 4$, as desired.

Example 3.7 (SMT Team 2024/8)**.** Find the number of ordered pairs (a, b) of positive integers such that $a^2 + b^2 \mid 2024^2$.

We'll first need the following claim, which is sometimes known as Fermat's Christmas theorem.

Claim. If p is a prime and $p \mid a^2 + b^2$, then either $p = 2$, $p \equiv 1 \pmod 4$, or $p \mid a, b$.

Proof. Suppose a, b are relatively prime and p is an odd prime dividing $a^2 + b^2$. We show $p \equiv 1 \pmod 4$, which is enough. The condition rearranges to $a^2 + b^2 \equiv 0 \pmod p$, or $(ab^{-1})^2 + 1 \equiv 0 \pmod p$, where b^{-1} is the multiplicative inverse of $b \pmod p$. But by the $n^2 + 1$ lemma this implies $p \equiv 1 \pmod 4$, and we are done. □

Since $2024^2 = 2^6 \cdot 11^2 \cdot 23^2$, the only primes dividing $a^2 + b^2$ are $2, 11, 23$, so there are no $1 \pmod 4$ primes. If a and b are relatively prime, the only possible pairs are $(1, 1)$, $(2, 2)$, and $(4, 4)$. You can multiply these pairs by 1, 11, 23, and $11 \cdot 23$ for a total of $4 \cdot 3 = 12$ pairs.

Example 3.8. How many primitive roots are there $\pmod p$?

Translating the problem into $\mathbb{Z}/(p-1)\mathbb{Z}$, we want to count the $a \in \mathbb{Z}/(p-1)\mathbb{Z}$ such that $a, 2a, \dots, (p-2)a$ are nonzero $\pmod{p-1}$. This occurs iff a is relatively prime to $p-1$, so there are $\varphi(p-1)$ primitive roots.

Exercise 3.9. For each $d \mid p-1$, prove there are $\varphi(d)$ elements of order d.

4 The composite case

Groups that are isomorphic to $(\mathbb{Z}/n\mathbb{Z}, +)$ for some n are called *cyclic*. We saw previously that $((\mathbb{Z}/p\mathbb{Z})^\times, \cdot)$ is cyclic. It turns out there exist primitive roots modulo n iff n is of the form $2, 4, p^k, 2p^k$ where p is an odd prime and k is a positive integer. In other words, $(\mathbb{Z}/n\mathbb{Z})^\times$ is cyclic, or,

$$((\mathbb{Z}/n\mathbb{Z})^\times, \cdot) \cong (\mathbb{Z}/\varphi(n)\mathbb{Z}, +).$$

Here, the idea of a primitive root is generalized to mean

$$g^1, g^2, \dots, g^{\varphi(n)-1}$$

are all not congruent to $1 \pmod n$.

Example 4.1 (Weak HMMT Guts 2019/28)**.** For how many integers a with $0 \le a \le 24$ is the sequence $a^1, a^2, a^4, a^8, \dots$ eventually constant $\pmod{25}$?

Observe the 5 multiples of 5 all work. Now we consider the residue classes relatively prime to 25. Since $\varphi(25) = 20$, we see $(\mathbb{Z}/25\mathbb{Z})^\times \cong \mathbb{Z}/20\mathbb{Z}$. Therefore, we want the number of $a \in \mathbb{Z}/20\mathbb{Z}$ such that the sequence $a, 2a, 4a, 8a, \dots$ is eventually constant modulo 20. This is just the multiples of 5, of which there are 4. Therefore, the answer is 9.

We've seen the structure of $((\mathbb{Z}/p^k\mathbb{Z})^\times, \cdot)$ is remarkably simple. However, we still don't really know how to deal with general n with multiple distinct prime factors. However, the *Chinese remainder theorem* allows us to *split* general n into its component prime powers, then deal with those prime powers.

Theorem 4.2 (Chinese remainder theorem)**.** For coprime n_1 and n_2, every ordered pair of residues

$$(r_1 \pmod{n_1}, r_2 \pmod{n_2})$$

corresponds *uniquely* to a residue $r \pmod{n_1 n_2}$.[1]

Intuitively, this means prime powers *don't interact*. I think the only way you can really get a feel for this theorem is doing problems with it. Let's see a few examples:

Example 4.3 (CMIMC ANT 2023/3)**.** Compute

$$2022^{\left(2022^{\cdot^{\cdot^{2022}}}\right)} \pmod{111}$$

where there are 2022 2022s.

Let x_n denote a power tower composed of n copies of 2022, we want to compute x_{2022} modulo 111. Since $111 = 3 \cdot 37$, by the Chinese remainder theorem, it's enough to compute the expression modulo 3 and 37. Since $3 \mid 2022$, $3 \mid x_{2022}$, so $x_{2022} \equiv 0 \pmod 3$. Now since $6 \mid 2022$, $36 \mid x_{2021}$, so by Fermat's little theorem, $x_{2022} \equiv 1 \pmod{37}$.

So now we must determine $x_{2022} \pmod{111}$. We know x_{2022} is of the form $37k+1$, and we want this to be $0 \pmod 3$. In other words, we want $37k+1 \equiv k+1 \equiv 0 \pmod 3$, so $k = 2 \pmod 3$, so $x_{2022} \equiv 37 \cdot 2 + 1 \equiv 75 \pmod{111}$.

Example 4.4 (HMMT Guts 2019/28)**.** How many positive integers $2 \le a \le 101$ have the property that there exists a positive integer N for which the last two digits in the decimal representation of a^{2^n} is the same for all $n \ge N$?

This is the same as the earlier problems but asking for 100 instead of 41 or 25. The key insight here is that we can split $\mathbb{Z}/100\mathbb{Z}$ into $\mathbb{Z}/4\mathbb{Z}$ and $\mathbb{Z}/25\mathbb{Z}$ by Chinese remainder theorem:

- In $\mathbb{Z}/25\mathbb{Z}$, there are 9 working residues by Example 4.1.
- In $\mathbb{Z}/4\mathbb{Z}$, clearly all 4 residues work.

Each pair of working residues in $\mathbb{Z}/25\mathbb{Z}$ and $\mathbb{Z}/4\mathbb{Z}$ uniquely correspond to a working residue in $\mathbb{Z}/100\mathbb{Z}$, again by Chinese remainder theorem, so the answer is $9 \cdot 4 = 36$.

Example 4.5 (OTIS Mock AIME 2024/13)**.** Compute the sum of all integers n such that $1 \le n \le 2024$ and exactly one of $n^{22} - 1$ and $n^{40} - 1$ is divisible by 2024.

Let a, b, and c be the number of n such that $n^{22} \equiv 1 \pmod{2024}$, $n^{40} \equiv 1 \pmod{2024}$, and $n^2 \equiv 1 \pmod{2024}$, respectively. Notice that if n works, so does $-n$, so we just want 1012 times the number of n that work, or $1012(a+b-2c)$ by PIE.

We study the structure of the group $(\mathbb{Z}/2024\mathbb{Z})^\times$ under multiplication. By Chinese remainder theorem, the fact that $(\mathbb{Z}/8\mathbb{Z})^\times = (\mathbb{Z}/2\mathbb{Z})^2$, and $(\mathbb{Z}/p\mathbb{Z})^\times \cong (\mathbb{Z}/(p-1)\mathbb{Z})$, we compute

$$(\mathbb{Z}/2024\mathbb{Z})^\times \cong (\mathbb{Z}/8\mathbb{Z})^\times \times (\mathbb{Z}/11\mathbb{Z})^\times \times (\mathbb{Z}/23\mathbb{Z})^\times$$

[1] In abstract algebra terms, we have the ring isomorphism $\mathbb{Z}/n_1n_2\mathbb{Z} \cong \mathbb{Z}/n_1\mathbb{Z} \times \mathbb{Z}/n_2\mathbb{Z}$.

$$\begin{aligned}&\cong (\mathbb{Z}/2\mathbb{Z})^2 \times \mathbb{Z}/10\mathbb{Z} \times \mathbb{Z}/22\mathbb{Z}\\&\cong (\mathbb{Z}/2\mathbb{Z})^2 \times \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/11\mathbb{Z}\\&\cong (\mathbb{Z}/2\mathbb{Z})^4 \times \mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/11\mathbb{Z}.\end{aligned}$$

Let this set be S. Observe a is the number of elements $s \in S$ such that $22s = (0,0,0,0,0,0)$, so counting component-wise shows $a = 2^4 \cdot 1 \cdot 11 = 176$. Similarly, b is the number of elements such that $40s = (0,0,0,0,0,0)$, so $b = 2^4 \cdot 5 \cdot 1 = 80$. Finally, c is the number of elements such that $2s = (0,0,0,0,0,0)$, so $c = 2^4 \cdot 1 \cdot 1 = 16$, so the answer is $1012(176 + 80 - 2 \cdot 16) = 226688$.

Example 4.6 (PUMaC NT 2016/7)**.** For how many integers n with $2017 \leq n \leq 2017^2$ do we have $n^n \equiv 1 \pmod{2017}$?

Observe $p = 2017$ is prime. Observe the value of $n^n \pmod{p}$ is uniquely determined by $n \pmod{p}$ and $n \pmod{p-1}$, and that n takes on each ordered pair $(r_1, r_2) \pmod{p-1, p}$ exactly once. Therefore we just want the number of ordered pairs $(a, b) \in \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/(p-1)\mathbb{Z}$ such that $a^b \equiv 1 \pmod{p}$.

By Exercise 3.9, there are $\varphi(d)$ elements of order d for each $d \mid n$. For each element $a \in (\mathbb{Z}/p\mathbb{Z})^\times$, there are $\frac{p}{\operatorname{ord}_p(a)}$ possible b such that $a^b \equiv 1 \pmod{p}$. Summing over a, our answer is

$$\sum_{a=0}^{p-1} \frac{p-1}{\operatorname{ord}_p(a)}.$$

By casework on $\operatorname{ord}_p(a)$, this becomes

$$\sum_{d \mid p-1} \varphi(d) \frac{p-1}{d}.$$

Note this sum is just $(\varphi * \mathrm{id})(p-1)$, where $*$ is the Dirichlet convolution. Since $\varphi * \mathrm{id}$ is multiplicative, the sum is just

$$\left(2^5 + 5(2^5 - 2^4)\right)\left(3^2 + 2(3^2 - 3)\right)(7 + 7 - 1) = 30576.$$